

Digital Logic Design

Practice Problem Set

- On the intersection of a main highway with the secondary access road, vehicle detection sensors are placed along lanes C and D (main road) and A and B (access road). These sensor outputs are LOW(0) when no vehicle is present and HIGH(1) when a vehicle is present. The intersection traffic is to be controlled by two traffic lights, N-S and E-W, using the following logic:
 - The E-W (east-west) traffic will be green whenever both lanes C and D are occupied.
 - The E-W light will be green whenever either C or D is occupied but lanes A and B are both not occupied.
 - The N-S (north-south) light will be green whenever both lanes A and B are occupied but C and D are both not occupied.
 - The N-S light will also be green when either A or B is occupied while C and D are both vacant.
 - The E-W light will be green when no vehicle is present.

Using the sensor outputs A, B, C and D as inputs, design a logic circuit to control the traffic lights. There should be two outputs N-S and E-W, which go HIGH(1) when the corresponding light is to be green. Simplify the circuit as much as possible.

- An air-conditioning unit is controlled by four variables: Temperature T, humidity H, the time of the day D, and the day of the week W. The unit is turned on under any of the following circumstances:
 - The temperature exceeds 78° F, and the time of the day is between 8 a.m. and 5 p.m.
 - The humidity exceeds 85%, the temperature exceeds 78° F, and the time of the day is between 8 a.m. and 5 p.m.
 - The humidity exceeds 85%, the temperature exceeds 78° F, and it is a weekend.
 - It is Saturday or Sunday and humidity exceeds 85%.

Write a logic expression for controlling the air-conditioning unit. Simplify the expression obtained as far as possible.

- Design a combinational circuit with three inputs, x, y and z, and three outputs, A, B and C. When the binary input is 0, 1, 2 or 3, the binary output is one greater than the input. When the binary input is 4, 5, 6 or 7, the binary output is one less than the input.

- A lawn sprinkling system at a large estate is to be automatically controlled by certain combinations of the following variables:

Season ($S = 1$ if summer, 0 otherwise)

Moisture content of soil ($M = 1$ if high, 0 if low)

Outside temperature ($T = 1$ if high, 0 if low)

Outside humidity ($H = 1$ if high, 0 if low)

The landscape architect designing the system wants the sprinkler to be turned ON under any of the following circumstances:

- Moisture content is low in the winter; or
- Temperature is high and moisture content is low in the summer; or
- Temperature is high and humidity is high in the summer;
- It is impossible that temperature is low in the summer.

Use a Karnaugh map to find the simplest possible logic expression for turning ON the sprinkler system.

- A digital computer has five output devices connected to it as follows:

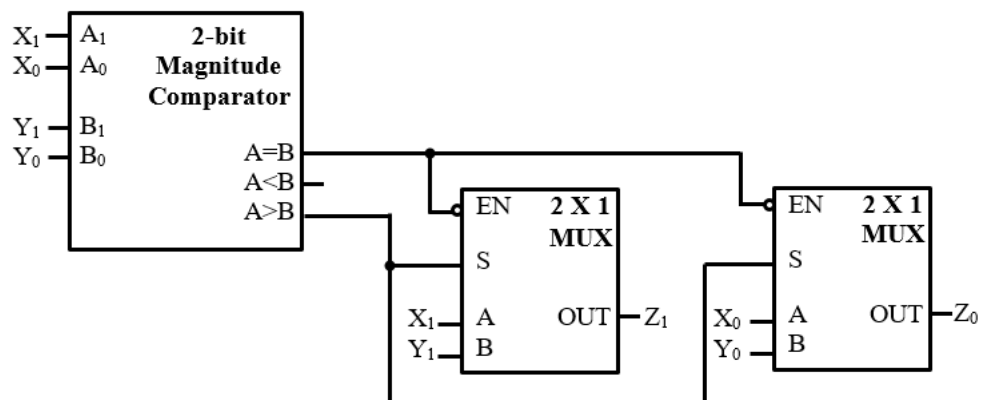
<u>Mode description</u>	<u>Code</u>	<u>Mode description</u>	<u>Code</u>
Punched card	001	Punched tape, numeric	100
Magnetic tape	010	Punched tape, alphabetic	101
Type writer	011		

The selection of an output device is controlled by signals on three parallel lines, denoted x, y and z, according to the codes indicated above. Design a combinational circuit that gives 1 at its output whenever punched tape is selected.

- Design half and full subtractors.

14. Design a 4x16 decoder (give gate level implementation).
15. Cascade two 4x16 decoders to form a 5x32 decoder. You may use any additional gate(s) if required.
16. Use five 2x4 decoders to form 4x16 decoder.
17. Design a BCD to 7-segment converter (give gate level implementation).
18. Design a hex to 7-segment converter (give gate level implementation).
19. Implement BCD to 7-segment decoder using a 4x16 decoder and OR gates.
20. Design (logic circuit) a 4x1 multiplexer having a single enable input line. All input and output lines are required to be active high.
21. Show how a 1 x 8 demultiplexer can be designed at gate level.
22. Cascade two 16x1 multiplexers to form a 32x1 multiplexer. You may use any additional gate(s) if required.
23. Use five 4x1 multiplexers to form a 16x1 multiplexer.
24. Consider an octal to binary priority encoder with priority scheme: higher digit higher priority. Octal inputs and binary outputs are all active high. The encoder has an active low enable input and two active low outputs E0 and GS such that E0 is active when none of the inputs are ON and GS is active when any of the inputs is ON.
 - i. Tabulate truth table including all inputs and outputs stated above.
 - ii. Derive equation for the least significant line of the binary output. Show all simplification steps.
25. Consider three 4-bit numbers: $A = A_3A_2A_1A_0$, $B = B_3B_2B_1B_0$ and $C = C_3C_2C_1C_0$. Show how two 7485 magnitude comparators can be connected to compare these three 4-bit numbers to find out that $A < B < C$. Use any external gates if required.
26. Consider the logic circuit shown in the figure. The magnitude comparator compares two 2-bit values and produces their comparative results on the three outputs. Each 2x1 multiplexer shown has two data inputs A and B, one select input S and an active low enable. OUT represents the output of the multiplexer. Compute the binary values at the outputs Z_1Z_0 if:

- i. $X_1X_0 = 10$ and $Y_1Y_0 = 01$
- ii. $X_1X_0 = 10$ and $Y_1Y_0 = 11$
- iii. $X_1X_0 = 11$ and $Y_1Y_0 = 11$



27. Consider the circuit given in the figure.
- Determine the Boolean function that the multiplexer implements.
 - Simplify the Boolean function using k-map.

